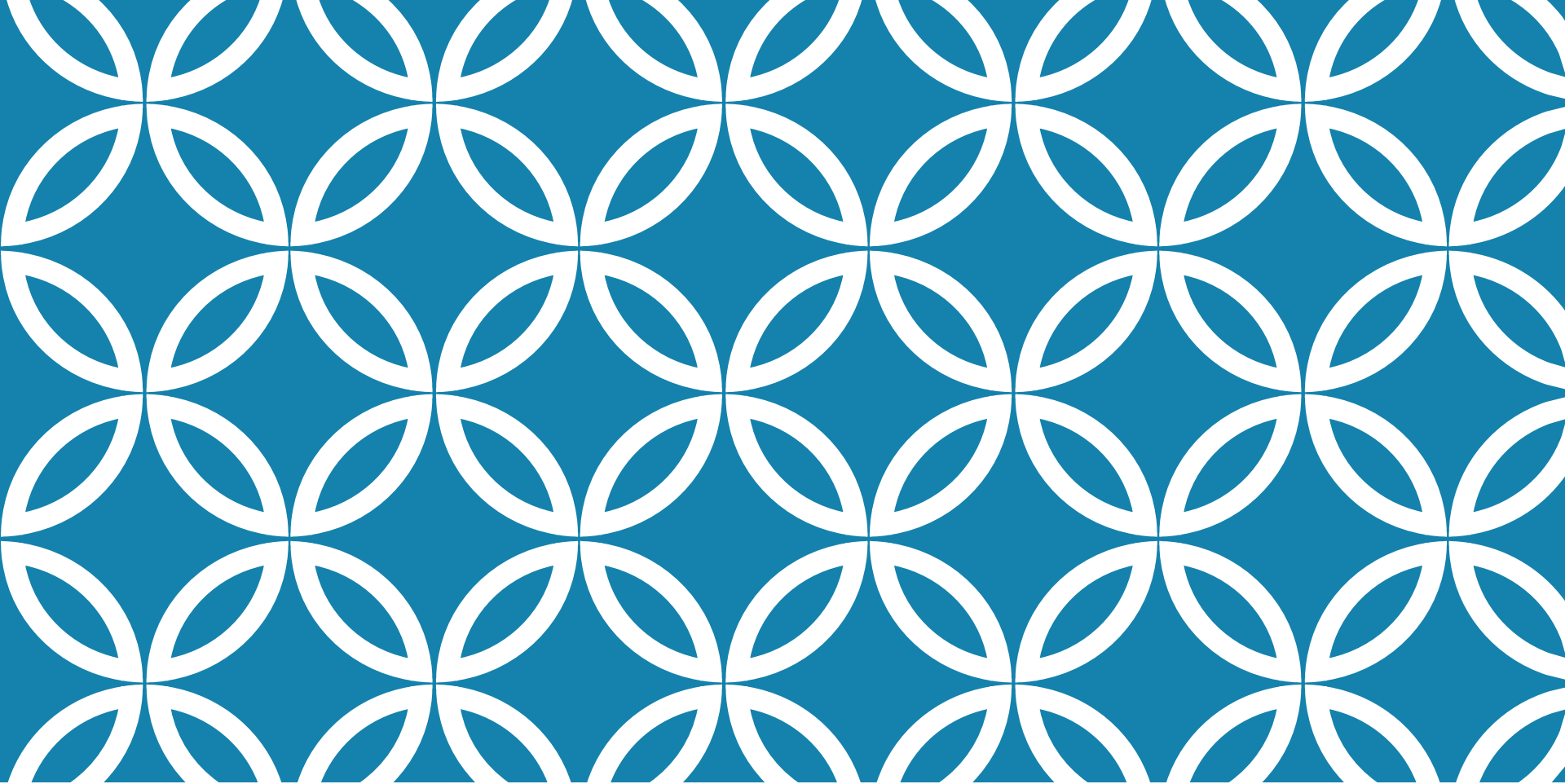




INTRODUCTION TO PROPOSITIONAL LOGIC AND PREDICATE LOGIC

INTRODUCTION



What is Logic?



- Definition and importance



- Types of logic:
Propositional and
Predicate

LOGIC

Logic is the study of correct reasoning.

It includes both formal and informal logic.

Formal logic is the study of deductively valid inferences or logical truths. It examines how conclusions follow from premises based on the structure of arguments alone, independent of their topic and content.

Informal logic is associated with informal fallacies, critical thinking, and argumentation theory. Informal logic examines arguments expressed in natural language whereas formal logic uses formal language.

LOGIC

Logical reasoning is one of the main subjects of computer science and mathematics. This reasoning helps in making the decisions of whether different types of mathematical arguments are correct or not.

Propositional logic and predicate logic are foundational elements of formal logic, which is essential for the study of mathematics, computer science, and philosophy. Both systems provide frameworks for understanding and constructing valid arguments, though they operate at different levels of complexity.

WHAT IS PROPOSITIONAL LOGIC?

A **proposition** is basically a declarative sentence that has a truth value.

Truth value can either be true or false, but it needs to be assigned any of the two values and not be ambiguous. The purpose of using propositional logic is to analyze a statement, individually or compositely.

PROPOSITIONAL LOGIC

Definition

- Deals with propositions and their truth values

Basic Components

- Propositions
- Logical connectives (AND, OR, NOT, IF...THEN, IF AND ONLY IF)
- t and f

A tautology is a compound statement that is true for every value of the individual statements.

LOGICAL CONNECTIVES

- AND ($\wedge$)-conjunction
- OR ($\vee$)- disjunction
- NOT ($\neg$)
- IMPLIES ($\rightarrow$) ($\neg P \vee Q$)
- IF AND ONLY IF ($\leftrightarrow$) ($P \rightarrow Q$) $\wedge$ ($Q \rightarrow P$)

EXAMPLES OF PROPOSITIONS

Simple Propositions

- Examples: 'It is raining', 'The sky is blue'

Compound Propositions

- Examples: 'It is raining AND the sky is blue'

TRUTH TABLES

Purpose

- To determine the truth value of propositions

Example

- Truth table for AND, OR, NOT

TRUTH TABLE

P	Q	$P \vee Q$	$P \wedge Q$
F	F	F	F
F	T	T	F
T	F	T	F
T	T	T	T

LAWS

Equivalence	Law
$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity Laws
$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination Laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent Laws
$\neg(\neg p) \equiv p$	Double Negation Law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative Laws

VIDEO LINKS

<https://www.youtube.com/watch?v=5NGKbiA04Cw&t=559s>

QUESTIONS

1. (30 pt.) Use the laws of propositional logic to prove that the following compound propositions are logically equivalent.
- a. (5 pt.) $(p \vee q) \wedge \neg(p \wedge \neg q)$ and q
 - b. (5 pt.) $\neg p \rightarrow \neg(q \vee r)$ and $(q \rightarrow p) \wedge (r \rightarrow p)$
 - c. (10 pt.) $\neg(p \vee (q \wedge (\neg p \rightarrow \neg r)))$ and $\neg p \wedge (q \rightarrow r)$
 - d. (10 pt.) $p \leftrightarrow (p \wedge q)$ and $\neg p \vee q$
-

WHAT IS PREDICATE LOGIC?

Predicates are properties, additional information to better express the subject of the sentence. A quantified predicate is a proposition, that is, when you assign values to a predicate with variables it can be made a proposition.

FIRST-ORDER LOGIC

First-order logic—also called predicate logic, predicate calculus, quantificational logic—is a collection of formal systems used in mathematics, philosophy, linguistics, and computer science.

First-order logic uses quantified variables over non-logical objects, and allows the use of sentences that contain variables.

Rather than propositions such as "all men are mortal", in first-order logic one can have expressions in the form "for all x , if x is a man, then x is mortal"; where

"for all x " is a quantifier,

x is a variable, and

"... *is a man*" and "... *is mortal*" are predicates.

PREDICATE LOGIC

Definition

- Extends propositional logic by dealing with predicates and quantifiers

Basic Components

- Predicates
- Quantifiers (Universal $\forall$, Existential $\exists$)

EXAMPLES OF PREDICATES

Simple Predicates

- Examples: 'x is a student', 'y is a teacher'

Quantified Statements

- Examples: 'For all x, x is a student', 'There exists a y such that y is a teacher'

APPLICATIONS

Propositional Logic

- Used in digital circuits, software engineering

Predicate Logic

- Used in databases, artificial intelligence

QUANTIFIERS

Quantifiers: Quantifiers specify the scope of variables. The two main quantifiers are:

Universal Quantifier ($\forall$): Indicates that a predicate applies to all elements in the domain.

For example, $\forall x (Person(x) \rightarrow Mortal(x))$ means "All persons are mortal."

Existential Quantifier ($\exists$): Indicates that there is at least one element in the domain for which the predicate holds.

For example, $\exists x (Person(x) \wedge Likes(x, IceCream))$ means "There exists a person who likes ice cream."

QUANTIFICATION

Quantifiers express the extent to which a predicate is true over a range of elements.

<https://calcworkshop.com/logic/predicate-logic/>

USING FIRST-ORDER LOGIC FOR REASONING

To illustrate the use of FOL in reasoning, consider the following knowledge base:

$\forall x (\text{Cat}(x) \rightarrow \text{Mammal}(x))$ (All cats are mammals)

$\forall x (\text{Mammal}(x) \rightarrow \text{Animal}(x))$ (All mammals are animals)

$\text{Cat}(\text{Tom})$ (Tom is a cat)

From these statements, we can infer:

$\text{Mammal}(\text{Tom})$ (Since Tom is a cat, and all cats are mammals)

$\text{Animal}(\text{Tom})$ (Since Tom is a mammal, and all mammals are animals)

VIDEO LINKS

<https://www.youtube.com/watch?v=tFYngFwDuJQ&t=401s>

<https://www.youtube.com/watch?v=Z0vTGof02n8>

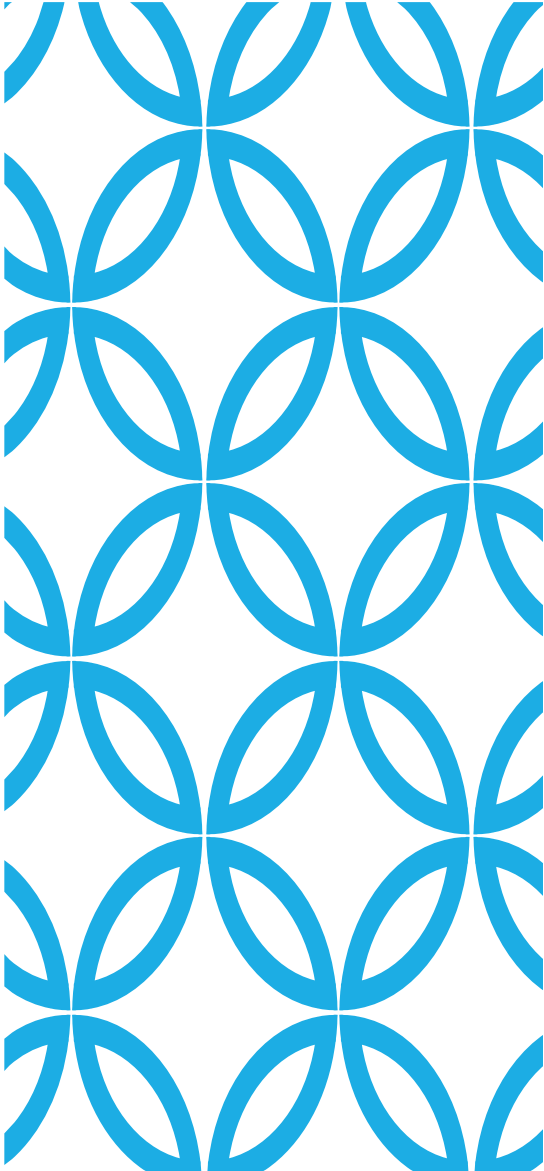
CONCLUSION

Summary

- Recap of key points

Importance of Logic

- In various fields of study and real-world applications



RESEARCH TOPIC

Difference between
predicate and propositional
logic.



IN-CLASS ASSIGNMENT

<https://www.geeksforgeeks.org/difference-between-propositional-logic-and-predicate-logic/>

<https://www.tutorialspoint.com/difference-between-propositional-logic-and-predicate-logic>